

AI Audit Report

1. Declaration

I use AI tools for the following tasks.

No.	AI Tool	Purpose
1	Codex	Assist in generating, formatting, auditing, and exporting HW02 Domain Testing and Boundary Value Analysis artifacts.

2. Instructions Applied

- Add one row per AI-generated artifact.
- Paste the verbatim prompt. Do not paraphrase.
- Paste the verbatim AI output, or identify the generated artifact when the output is too large to reproduce inside the table.
- Tag the verdict as `VALID`, `INVALID`, or `INCOMPLETE`.
- Reasoning cites an ISTQB Foundation Level section, assignment/course instruction, or technical RFC.
- Show the corrected artifact in the Student Fix column with the changed part highlighted in bold.

3. Audit Table - One Row Per Artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB / RFC)	(5) Student Fix
Artifact: <code>HW02/test-cases/FR17-coupon-management-crud.md</code> Tool: Codex Time: 2026-06-22 13:45 +07 Prompt: "Use the domain-boundary-testing skill for FR-17 Coupon Management CRUD. Goal: Create the first draft of testing artifacts for this feature. Requirements: identify inputs/outputs, validation rules and assumptions, equivalence classes, Domain Testing test cases, Boundary Value Analysis test cases, AI Gap Analysis, Human Review notes, and suggest a Git commit message. Do not invent actual execution results. Use Actual Result = TBD and Status = Not Executed. Save the output to <code>test-cases/FR17-coupon-management-crud.md</code> ."	Generated output: <code>HW02/test-cases/FR17-coupon-management-crud.md</code> initial draft with Feature Understanding, Input/Output Identification, Domain Rules, Equivalence Classes, Domain Testing cases, BVA cases, execution table, bug template, AI Gap Analysis, Human Review Notes, and suggested commit message.	IN CO MP LE TE	ISTQB FL §4.2 (Black-box Test Techniques) requires identifying valid and invalid partitions and boundary values. The first AI draft covered common required-field and numeric boundaries but missed input-normalization partitions for coupon code.	Added <code>FR17-EC-029</code> to <code>FR17-EC-032</code> , <code>FR17-DT-018</code> to <code>FR17-DT-021</code> , and <code>FR17-BVA-022</code> for whitespace-only code, leading/trailing spaces, separator characters, Unicode code, and max-length behavior. Changed unclear accepted code-format cases to <code>Inconclusive – Requirement unclear</code> instead of confirmed defects.
Artifact: <code>HW02/test-cases/FR17-coupon-management-crud.md</code> and <code>HW02/reports/bug-report.md</code> Tool: Codex Time: 2026-06-23 12:56 +07 Prompt: "Dựa vào kết quả các test case hãy viết lại phần 8 và <code>bug-report.md</code> dựa trên skill đồng thời dùng attach hình ảnh tôi đã cung cấp trong folder screenshot."	Generated output: updated FR-17 execution summary and bug-report groups based on executed failures and screenshots.	VA LID	ISTQB FL §5.6 (Defect Management) expects failed test results to be recorded with expected result, actual result, evidence, and traceability to the related test case.	Accepted grouping of failed FR-17 validation cases into bug reports. Updated Section 8 totals to match executed tests and linked available screenshot evidence in <code>bug-report.md</code> .
Artifact: <code>HW02/test-cases/FR09-discount-coupons.md</code> Tool: Codex	Generated output: <code>HW02/test-cases/FR09-discount-coupons.md</code> with rules C1-C5, percent/fixed discount cases,	IN CO MP	ISTQB FL §4.2.3 (Boundary Value Analysis) requires testing just below, exactly at, and just above thresholds. AI	Updated executed results for FR09. Added defects for percent discount final amount unchanged, fixed discount greater than cart

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB / RFC)	(5) Student Fix
Time: 2026-06-23 16:01 +07 Prompt: "Use the domain-boundary-testing skill for FR-09 Discount Coupons. Create test cases for valid coupon, invalid coupon, expired coupon, inactive coupon, usage limit reached, cart total below minimum order, percentage discount, fixed amount discount, and discount greater than cart total. Do not invent actual execution results. Save the output to test-cases/FR09-discount-coupons.md."	minimum-order cases, usage-limit cases, BVA cases, and AI Gap Analysis.	LE TE	designed threshold cases but could not verify usage history, discount calculation, or lockout-like state without execution data.	total producing negative value, and exact minimum-order rejection. Clarified rounding and expiration-date semantics as assumptions to verify.
Artifact: HW02/test-cases/FR05-product-listing-search.md Tool: Codex Time: 2026-07-01 11:40 +07 Prompt: "Using domain-boundary-testing skill for HW02/test-cases/FR05-product-listing-search.md. INFORMATION: Trang chủ hiển thị danh sách tất cả sản phẩm dạng lưới (grid). Mỗi sản phẩm hiển thị: Ảnh (tỷ lệ chuẩn, có alt text mô tả), Tên sản phẩm, Giá (đơn vị: đ, định dạng phân cách hàng nghìn). Thanh tìm kiếm tìm theo tên sản phẩm. Từ khóa tìm kiếm phải được hiển thị an toàn (không render HTML). Khi đang tải dữ liệu phải hiển thị trạng thái loading. Khi không có kết quả tìm kiếm phải hiển thị thông báo empty state phù hợp."	Generated output: HW02/test-cases/FR05-product-listing-search.md with product grid/search partitions, safe keyword handling, keyword length BVA, and product-count BVA.	IN CO MP LE TE	ISTQB FL §1.4 (Test Activities) and §4.2 (Test Techniques) require tests to be feasible in the selected test basis and environment. AI included image, invalid price, loading-state, and controlled product-count cases that were not executable in the available manual setup.	Removed unsuitable execution rows. Kept controlled product-count boundaries as Not Executed. Recounted Section 8 to 15 designed, 12 executed, 12 passed, 0 failed, 3 not executed.
Artifact: HW02/test-cases/Mobile-FR04-profile-management.md Tool: Codex Time: 2026-07-02 18:57 +07 Prompt: "I have updated domain test case and boundary value analysis test case result of HW02/test-cases/Mobile-FR04-profile-management.md. Requirement: update Boundary Identification and Equivalence Class Partitioning to suitable to result; compare it with the last version to update 8.Test Execution Table, 10. AI Gap Analysis and 11. Human review note; format file md."	Generated output: updated Mobile-FR04 equivalence classes, boundary identification, execution totals, AI Gap Analysis, and Human Review Notes.	IN CO MP LE TE	ISTQB FL §1.4 (Test Activities) emphasizes comparing actual results against expected results and updating test analysis when execution reveals new behavior. AI initially trusted the stated phone rule and missed observed 9-10 digit behavior and order-history scope.	Changed phone partitions to reflect executed behavior: accepted/rejected values around 8, 9, 10, and 11 digits. Added order-history domain and BVA coverage for empty, one, and multiple orders, order fields, total formatting, status, and cancel action.
Artifact: HW02/test-cases/FR02-login-and-account-logout.md Tool: Codex Time: 2026-07-05 21:39 +07 Prompt: "Based on HW02/.agents/skills/domain-boundary-testing implement HW02/test-cases/FR02-login-and-account-logout.md \n\nINFORMATION: \n- Người dùng nhập Email và Mật khẩu.\n- Sau mỗi lần đăng nhập sai, hệ thống tăng bộ đếm lên đúng 1 đơn vị.\n- Nếu đăng nhập sai từ 3 lần trở lên liên tiếp, tài khoản bị tạm khóa 30 giây (môi trường demo). Hệ thống trả về thông báo lỗi phù hợp; không để lộ chi tiết nguyên nhân.\nImage: [Image #1] [Image #2]"	Generated output: HW02/test-cases/FR02-login-and-account-logout.md with 29 designed tests covering login, failed counter increment, 3-attempt lockout, 30-second lockout duration, generic error disclosure, and logout.	IN CO MP LE TE	RFC 5322 defines email syntax concerns, and ISTQB FL §4.2.3 (Boundary Value Analysis) requires testing 2, 3, and 4 failed attempts and 29, 30, and 31 seconds around the lockout boundary. The AI could only design tests because no execution evidence or backend counter access was provided.	Kept all FR02 cases as Actual Result = TBD and Status = Not Executed. Highlighted assumptions: UI field says Username while requirement says Email; login page title says Đăng Ký; failed counter needs DB/API visibility; exact 30-second unlock semantics must be verified.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB / RFC)	(5) Student Fix
Artifact: HW02/reports/main-report.md and HW02/reports/main-report.pdf Tool: Codex Time: 2026-07-05 11:31 +07 Prompt: "Phần main report bao gồm toàn bộ nội dung từ phần 1 đến phần 7 của từng FR có trong HW02/test-cases (ngoại trừ Assumptions to verify) chứ không phải tóm tắt như hiện tại. Hãy viết lại HW02/reports/main-report.md giúp tôi"	Generated output: HW02/reports/main-report.md rewritten to include full sections 1-7 from each feature file, excluding Assumptions to verify; later exported to HW02/reports/main-report.pdf.	VALID	Course main-report instruction requires the required Domain Testing and Boundary Value Analysis sections. ISTQB FL §1.4 (Test Work Products and Traceability) supports traceability from summary report back to detailed test cases.	Changed main report from summary-only to full section inclusion. After FR02 was added, added FR-02 to the report and updated totals to 160 designed, 128 executed, 78 passed, 41 failed, 32 not executed, 20 bugs.
Artifact: HW02/reports/ai-critique.md and HW02/reports/ai-critique.pdf Tool: Codex Time: 2026-07-05 11:53 +07 Prompt: "Dựa vào 10 AI Gap analysis của từng FR trong HW02/test-cases viết cho tôi 1 file AI Critique (200–300 words, Mandatory)\nWrite a paragraph of 200–300 words critiquing the AI. Address the following\nquestions: Where did the AI get something wrong, biased, or incomplete? Why\ndid it fail to catch the issue? What principle have you learned about\ncollaborating with AI during this assignment?"	Generated output: HW02/reports/ai-critique.md, one paragraph, 253 words; exported to HW02/reports/ai-critique.pdf.	VALID	Course AI Critique instruction requires a 200-300 word critique addressing AI mistakes, why they happened, and the collaboration lesson. The output is within the required word count and cites concrete AI Gap Analysis themes across FR05, FR09, FR17, and Mobile-FR04.	Accepted one-paragraph structure. Included concrete critique of unsuitable test cases, unclear assumptions, missed normalization, stale counts, and over-trust in stated rules.
Artifact: HW02/git-commit-log.txt Tool: Codex Time: 2026-07-05 11:57 +07 Prompt: "Sử dụng git log --graph --all --stat và đưa nội dung đó vào HW02/git-commit-log.txt"	Generated output: HW02/git-commit-log.txt, 367 lines from git log --graph --all --stat.	VALID	Course deliverable instruction requires submission traceability. This artifact is command output, so validity depends on preserving the exact command output rather than AI interpretation.	Used exact command git log --graph --all --stat > HW02/git-commit-log.txt. No AI paraphrase was added to the file.
Artifact: HW02/reports/ai-audit-report.md Tool: Codex Time: 2026-07-06 10:08 +07 Prompt: "[Image #1] Chỉnh sửa HW02/reports/ai-audit-report.md lại sao cho phù hợp với yêu cầu ảnh tôi cung cấp cũng như fill human review note giúp tôi, bằng cách so sánh những thay đổi sau khi tôi thao tác"	Generated output: this revised HW02/reports/ai-audit-report.md, converted from an interaction-log format to a five-column artifact audit table with prompt, AI output, verdict, ISTQB/RFC reasoning, and student fix.	VALID	Course AI Audit instruction screenshot requires one audit row per AI-generated artifact, a verdict, reasoned evaluation, and a corrected artifact/fix. This rewrite follows that structure.	Replaced old columns Interaction ID / AI Tool / Date / Task / User Prompt / AI Output Summary / Human Review with Prompt + Tool / AI Output / Verdict / Reasoning (ISTQB / RFC) / Student Fix. Filled human review notes from current file diffs and AI Gap Analysis records.

4. Human Review Notes

Human Review:
– Accepted:
– Codex was useful for drafting structured Domain Testing and Boundary Value Analysis artifacts.
– Generated reports and PDFs were accepted after render checks.

- The git commit log artifact was accepted because it used exact git command output.
- Modified:
 - Rewrote this AI Audit Report to match the required five-column table shown in the assignment image.
 - Replaced generic "Pending human review" values with concrete VALID / INCOMPLETE verdicts.
 - Added ISTQB/RFC-based reasoning to each audited artifact row.
 - Updated main report totals after adding FR-02: 160 designed, 128 executed, 78 passed, 41 failed, 32 not executed, 20 bugs.
- Removed:
 - Removed the old interaction-log-only structure because it did not show verdict, reasoning, and student fix per artifact.
 - Removed sample-style pending text where actual human fixes were already visible from the test-case files.
- Added:
 - Added FR-02 audit row for login, failed counter, 3-attempt lockout, 30-second lockout, and logout artifact.
 - Added explicit student fixes for FR05, FR09, FR17, Mobile-FR04, FR02, Main Report, AI Critique, and Git Commit Log.
 - Added comparison basis from current working tree: main report was modified to include FR-02; PDF outputs were regenerated; AI audit report was rewritten.
- Assumptions to verify:
 - Exact timestamps should be checked against final commit history if strict timestamp accuracy is required.
 - Full verbatim AI outputs are stored in the generated artifact files; the table includes artifact paths and concise output excerpts to keep the report readable.
 - Course evaluator may require screenshots of AI output instead of file references for very large generated artifacts.

5. Corrected Artifact Highlights

Artifact	Highlighted Correction
HW02/test-cases/FR17-coupon-management-crud.md	Added coupon-code normalization classes and tests for whitespace, trimming, separator, Unicode, and max-length behavior.
HW02/test-cases/FR09-discount-coupons.md	Recorded defects from execution for unchanged percent final amount, negative fixed-discount final amount, and exact-minimum rejection.
HW02/test-cases/FR05-product-listing-search.md	Removed unsuitable non-executable cases and corrected execution totals to match remaining tests.
HW02/test-cases/Mobile-FR04-profile-management.md	Updated phone boundary analysis from execution results and added order-history coverage.
HW02/test-cases/FR02-login-and-account-logout.md	Created draft FR02 test artifact with all cases marked Not Executed because screenshots are context only.
HW02/reports/main-report.md	Added FR-02 and updated overall totals after comparing current HW02/test-cases/ content.
HW02/reports/ai-critique.md	Replaced placeholder critique with a 253-word mandatory critique paragraph.
HW02/reports/ai-audit-report.md	Changed audit format to match the required one-row-per-artifact table.

6. Suggested Git Commit Message

Revise AI audit report to assignment table format